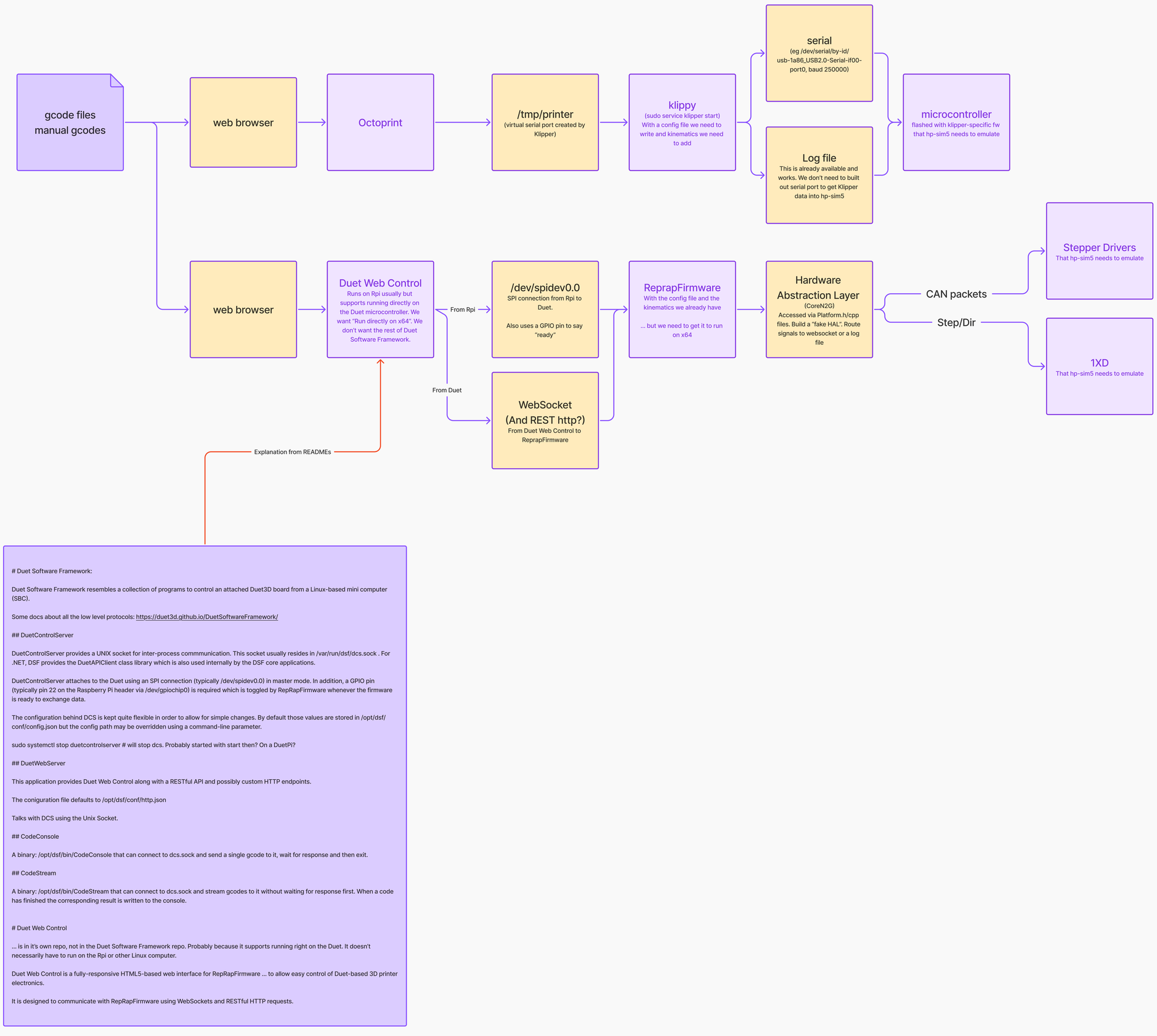
Flow chart



Duet Software Framework:

Duet Software Framework resembles a collection of programs to control an attached Duet3D board from a Linux-based mini computer (SBC).

Some docs about all the low level protocols: https://duet3d.github.io/DuetSoftwareFramework/

DuetControlServer

DuetControlServer provides a UNIX socket for inter-process commmunication. This socket usually resides in /var/run/dsf/dcs.sock . For .NET, DSF provides the DuetAPIClient class library which is also used internally by the DSF core applications.

DuetControlServer attaches to the Duet using an SPI connection (typically /dev/spidev0.0) in master mode. In addition, a GPIO pin (typically pin 22 on the Raspberry Pi header via /dev/gpiochip0) is required which is toggled by RepRapFirmware whenever the firmware is ready to exchange data.

The configuration behind DCS is kept quite flexible in order to allow for simple changes. By default those values are stored in /opt/dsf/conf/config.json but the config path may be overridden using a command-line parameter.

sudo systemctl stop duetcontrolserver # will stop dcs. Probably started with start then? On a DuetPi?

DuetWebServer

This application provides Duet Web Control along with a RESTful API and possibly custom HTTP endpoints.

The coniguration file defaults to /opt/dsf/conf/http.json

Talks with DCS using the Unix Socket.

CodeConsole

A binary: /opt/dsf/bin/CodeConsole that can connect to dcs.sock and send a single gcode to it, wait for response and then exit.

CodeStream

A binary: /opt/dsf/bin/CodeStream that can connect to dcs.sock and stream gcodes to it without waiting for response first. When a code has finished the corresponding result is written to the console.

Duet Web Control

... is in it's own repo, not in the Duet Software Framework repo. Probably because it supports running right on the Duet. It doesn't necessarily have to run on the Rpi or other Linux computer.

Duet Web Control is a fully-responsive HTML5-based web interface for RepRapFirmware ... to allow easy control of Duet-based 3D printer electronics.

It is designed to communicate with RepRapFirmware using WebSockets and RESTful HTTP requests.